

AutoFuzzer: Project & Firmware Specification

Embedded Communication Fuzzer & Target Firmware Robustness Testing

AutoFuzzer: Next Phase Hardware & Firmware Specification

1. AutoFuzzer Architecture & Next Phase Design

AutoFuzzer is an automated board-to-board hardware robustness testing platform. The next phase introduces interactive UI elements, real-time hardware status feedback, and expanded fuzzing control.

2. Hardware Components & Pin Layout

ESP32 Fuzzer & Controller Board Pinouts:

- OLED Display (SSD1306 128x64): SDA = GPIO 21, SCL = GPIO 22
- Status LED

3. Next Phase Firmware Features

A. Real-Time OLED Dashboard

- Live display of total packets sent, current mutation strategy, RNG seed, and target DUT status.
- Push button enables toggling between RUNNING, PAUSED, and REPLAY modes.
- Red LED illuminates and Buzzer sounds when target heartbeat drops below 350 ms threshold.
- Expansion from UART to SPI, I2C, and CAN bus protocols.

B. Interact

C. Audio-V

D. Multi-Bu